

Eight-Point Exact-Pressure Optimization for Simple Zeros of the Riemann Zeta Function

AMTOPA

August 12, 2026

Abstract

We record a computer-assisted research-draft refinement of the positioned-pressure method for simple zeros of the Riemann zeta function on the critical line. The current certified construction uses an eight-point local inequality, a re-optimized rational 15-term analytic window, exact shifted-pressure accounting, and a seven-dimensional outward-rounded interval certificate. Subject to the imported analytic interface and finite-dimensional Gram profile of the predecessor lineage, the certified numerical inputs yield

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T, 2T)}{N(T, 2T)} \geq 0.67340621629894866656 \dots,$$

i.e. 67.3406216299% before conservative decimal truncation. A further joint-pressure construction is being tested separately and is not included in this certified headline. This manuscript is a research draft, not a peer-reviewed theorem and not a proof of the Riemann hypothesis.

1 Imported analytic interface

Let $S = N_0^s(T, 2T)$ and $N = N(T, 2T)$. We import

$$S \geq HN + \Delta(M) - o(N)$$

and the finite- m Gram profile

$$h_m(E) = \begin{cases} E, & 0 \leq E \leq m/(m-1), \\ E/m + 2\sqrt{(m-1)E/m} - 1, & E \geq m/(m-1). \end{cases}$$

These ingredients belong to the Anthropic / `trmdy` / `sxuff` lineage and are not reproved here.

2 Certified analytic window

We use

$$v(s) = \sum_{j=0}^{14} c_j \cos(\omega_j s), \quad \omega_0 = \sqrt{2}, \quad \omega_j = 2j\pi \ (1 \leq j \leq 14),$$

with common denominator 10^9 and numerator vector

$$(1000000000, 7862237, -11036157, 3734450, 1402396, -1412889, 3402026, \\ -1739485, 4930665, -4255468, 3101956, -2601879, 146498, 398565, -271068).$$

Direct interval arithmetic encloses

$$H(v) = 0.672340545121386742008925898118525 \dots$$

and verifies

$$H_{\text{cert}} = 0.672340545 < H(v).$$

The same interval subdivision proves positivity of the window on the required support.

3 Eight-point local functional

Let $g_1, \dots, g_7 \geq 0$ and $y_0 = 0$, $y_j = g_1 + \dots + g_j$. We use

$$F_8(g) = \sum_{r=1}^7 b_r g_r + \sum_{0 \leq i < j \leq 7} a_{ij} W(y_j - y_i).$$

The exact pair weights are recorded in `candidate.json`. They are nonnegative and satisfy

$$\sum_{i=0}^{7-s} a_{i,i+s} = 2 \quad (1 \leq s \leq 7).$$

The position-pressure coefficients are

$$\frac{1}{2300000000}(474488, 829921, 1102531, 1186120, 1102531, 829921, 474488),$$

so

$$B = \sum_{r=1}^7 b_r = \frac{3}{1150}.$$

4 Seven-dimensional interval certificate

Window-basin min-max exchange followed by unscreened stress over the relevant resonance families found a lowest floating basin near

$$0.005516121966580929 \dots$$

We certify the smaller rational target

$$\varepsilon_8 = \frac{1103}{200000} = 0.005515.$$

The rigorous verifier uses 4000 interval cells per normalized unit, 50 decimal digits in the outward-rounded table builder, and C++ compiled with `-ffp-contract=off`. It reports

$$\begin{aligned} \text{nodes} &= 66686224, & \text{pruned} &= 33343136, \\ \text{splits} &= 33343088, & \text{convex} &= 27190260, \\ \text{tangent} &= 12043940, & \text{max depth} &= 86. \end{aligned}$$

The output is `VERIFIED=true`. Exact table hashes, workflow run 31598015611, artifact 9142888339, and its digest are stored in the repository metadata.

5 Exact shifted-pressure averaging

For an m -point block there are $m - 7$ translated eight-point windows. Set

$$A_m = \varepsilon_8(m - 7), \quad R_m = h_m(A_m), \quad \eta_m = \frac{R_m}{A_m}.$$

The span-capacity identities control the summed pair contribution by the predecessor block energy, while exact shifted averaging gives pressure charge $(m - 7)B$. Thus

$$\Delta(M) \geq \frac{R_m}{m} S - \frac{\eta_m B(m - 7)}{m} N - o(N),$$

and therefore

$$\boxed{\frac{S}{N} \geq \frac{m H_{\text{cert}} - \eta_m B(m - 7)}{m - R_m} - o(1).}$$

6 Final arithmetic

With

$$H_{\text{cert}} = 0.672340545, \quad \varepsilon_8 = 0.005515, \quad B = 3/1150,$$

the integer scan selects $m = 203$. Then

$$A_{203} = 1.08094,$$

$$R_{203} = 1.07956220302209765100385204312294\dots,$$

and

$$\eta_{203} = 0.99872537145641538938687812748436\dots$$

Hence

$$\frac{S}{N} \geq 0.67340621629894866656390422423548\dots - o(1),$$

and safely

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T, 2T)}{N(T, 2T)} > 0.6734062162.$$

7 Joint-pressure extension under test

The predecessor block argument uses the total position pressure symbolically as $B = \sum b_r$. The active `discovery_candidate.json` tests the further degree of freedom obtained by optimizing this total together with the window and local weights, while preserving nonnegative pressures and pair-span capacities. Its full local interval certificate is not assumed in the certified result above.

8 Trust boundary and status

The interval window bound, exact pair-span identities, exact pressure total, seven-dimensional local certificate, shifted-pressure bookkeeping, and final arithmetic are directly checkable from this repository. The explicit-formula / trace interface and finite- m Gram profile remain imported from prior work. Independent reproduction of the seven-dimensional certificate is requested; accordingly, the percentage should be read as an interval-certified research-draft candidate rather than a publication-ready theorem.

Reproducibility

The repository contains exact current parameters in `candidate.json`, verifier evidence in `certificates/`, arithmetic and interval code in `src/`, and superseded records in `archive/`. The repository follows a single-main workflow.